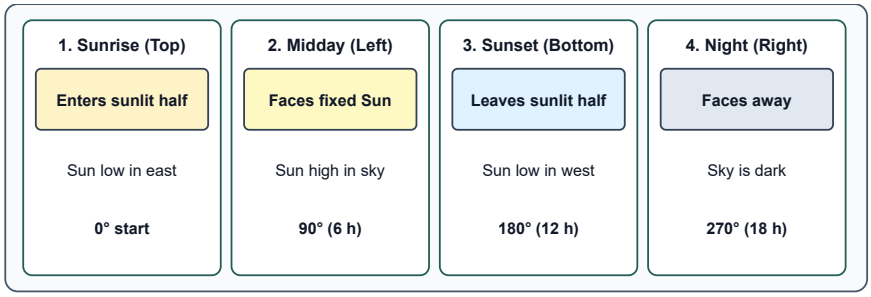
Science · GROW · Autumn 2 · Week 1 · GROW_SCI_A2_W01_P1

Knowledge organiser



Knowledge organiser model tracking one marked place through Sunrise, Midday, Sunset, and Night as Earth rotates.

Learning goal: I can explain how Earth's rotation causes day, night, and time zones, and how Earth's tilted axis causes changing day length and seasons.

Tier 2 and Tier 3 vocabulary

Tier	Word	Meaning	Example in this lesson
Tier 3	rotation	One full turn of Earth around its own axis, taking 24 hours (360°, or 15° of longitude per hour).	Earth's rotation carries our place from day into night and sets global time zones.
Tier 3	tilted axis	Earth's imaginary pole-to-pole spin line, tilted at 23.5° relative to its orbit around the Sun.	Earth's 23.5° tilted axis changes the length of day through the year and causes the seasons.
Tier 2	apparent	How a movement seems from an observer's viewpoint when the observer is moving with Earth.	Because we rotate west to east with Earth, the Sun has an apparent motion from east to west.
Tier 3	time zone	A region on Earth that keeps a shared local time based on 15° steps of longitude per hour of rotation.	Places 15° east see sunrise one hour earlier because Earth rotates west to east.
Tier 3	seasons	Yearly changes in daylight length and solar height caused by Earth's 23.5° tilted axis as Earth orbits the Sun.	When our hemisphere tilts towards the Sun we have summer with longer days; when it tilts away we have winter with shorter days.

Core Key Stage 3 science ideas: Earth rotates west to east on its axis once every 24 hours (anticlockwise from above the North Pole). The half facing the Sun has day and the half facing away has night. Because Earth turns 360° in 24 hours (15° per hour), global **time zones** shift by 1 hour every 15° of longitude. Because Earth's axis is **tilted at 23.5°** as it orbits the Sun, the **length of day changes through the year** and produces the **seasons** (summer when tilted towards the Sun; winter when tilted away; 12-hour day and night at the equinoxes).

Made by Matt